

ICWaR Course

WR 201

Title: Watershed Modeling (2:1)

Instructor: Dr. Rajarshi Das Bhowmik

Term: August - December

Course description: This course will cover the concepts of watershed modeling. This three- credit course will be offered as an elective every year in the August-December term in the ICWaR. This course is aimed to be an introductory graduate-level (200-series) course, typically with Water Resources Engineering background. In-class lectures include basic and advanced topics related to surface hydrology. Additionally, the basics of computer methods in hydrology will be discussed. Tutorial sessions on MATLAB, MS Excel, and ArcSWAT will be conducted. Calibration and validation of ArcSWAT (distributed model) and HYMOD (lumped model) will be performed during tutorial sessions.

Contents:

1. Introduction to watershed modeling

Runoff generation and streamflow. Spatio-temporal scales in watershed modeling. Watershed properties.

2. Pre-processing of model inputs

Understanding watershed model inputs. Basics of MATLAB computations. Hydrologic data processing in MATLAB.

3. Data-driven watershed models

Stochastic models (ex: Quantile regression). Neural network models.

4. Different watershed models

Lumped/ Distributed/ Physical/ Conceptual watershed models. Thornwaite-type water balance model. Thomas model (*abcd*) model. Data assimilation in *abcd* model. Introduction to ArcSWAT and HYMOD. Flood forecasting. Performance measures.

5. Predicting the future

Future hydrologic projections. Impact of climate change on river discharge. Uncertainty in river discharge estimation.

Tutorial Sessions

Processing of Geospatial and temporal data for watershed modeling. MATLAB tutorials. '*abcd*' model set-up in MS Excel. Stochastic and NN model setup in Matlab. Calibration and validation of HYMOD and ArcSWAT. Development of forecast models.

Prerequisites: None

Textbooks:

Dingman, S. L. (2015). *Physical hydrology*. Waveland press.

Singh, V. P., & Frevert, D. K. (Eds.). (2010). *Watershed models*. CRC press.